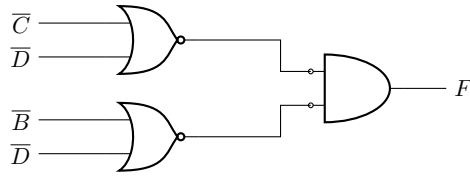


1. $F = \prod M(3, 5, 7, 11, 13, 15)$. Grouping the 0's: $\overline{F} = CD + BD$, so

$$F = (\overline{C} + \overline{D})(\overline{B} + \overline{D}).$$

		CD			
		00	01	11	10
AB	00	0	1	0	1
	01	1	0	0	1
	11	1	0	0	1
	10	1	1	0	1

NOR form (two faces of NOR: every OR $\rightarrow$ OR-INVERT, every AND $\rightarrow$ INVERT-AND):



2. $F(x, y, z) = \sum m(0, 1, 2, 5, 7)$.

$$F = \overline{x}\overline{z} + xz + \overline{y}z$$

		yz			
		00	01	11	10
x	0	1	1	0	1
	1	0	1	1	0

3. $F(A, B, C) = AB + \overline{A}\overline{C} + A\overline{B}C = \sum m(0, 2, 5, 6, 7)$.

$$F = \overline{A}\overline{C} + AB + AC$$

		yz			
		00	01	11	10
x	0	1	0	0	1
	1	0	1	1	1

(Using variable names A, B, C on the map: $x \rightarrow A, y \rightarrow B, z \rightarrow C$.)

4. $F(w, x, y, z) = \sum m(1, 3, 5, 7, 8, 9, 12, 13, 14, 15)$.

$$F = \overline{w}z + wx + w\overline{y}$$

		CD			
		00	01	11	10
AB	00	0	1	1	0
	01	0	1	1	0
	11	1	1	1	1
	10	1	1	0	0

(Map variables $w, x, y, z \rightarrow AB, CD$ as $A=w, B=x, C=y, D=z$.)

5. $F(A, B, C, D) = \sum m(0, 1, 2, 5, 6, 7, 8, 9, 10, 13, 14)$.

Prime implicants: $\overline{C}D, C\overline{D}, \overline{B}\overline{D}, \overline{B}\overline{C}, \overline{A}BD, \overline{A}BC$.

Essential: $\overline{C}D$ (only PI on m_5) and $C\overline{D}$ (only PI on m_2).

One minimum SOP:

$$F = \overline{C}D + C\overline{D} + \overline{B}\overline{C} + \overline{A}BC$$

		CD			
		00	01	11	10
AB	00	1	1	1	1
	01	0	1	1	1
	11	0	1	0	1
	10	1	1	0	1

6. $F(A, B, C, D) = \sum m(0, 1, 4, 5, 6, 7, 8, 10, 12, 14)$.

(a) SOP — group the 1's:

$$F = \overline{A}\overline{C} + \overline{A}B + A\overline{D}$$

		CD			
		00	01	11	10
AB	00	1	1	0	0
	01	1	1	1	1
	11	1	0	0	1
	10	1	0	0	1

(b) POS — group the 0's: $\overline{F} = AD + \overline{A}\overline{B}C$, so

$$F = (\overline{A} + \overline{D})(A + B + \overline{C}).$$

7. $F(x, y, z) = \prod M(1, 3, 5, 6)$. Grouping the 0's: $\overline{F} = \overline{x}z + \overline{y}z + xy\overline{z}$, so

$$F = (x + \overline{z})(y + \overline{z})(\overline{x} + \overline{y} + z).$$

		yz			
		00	01	11	10
x	0	1	0	0	1
	1	1	0	1	0

8. $F(w, x, y, z) = \sum m(0, 4, 6, 12, 14)$, $d = (2, 10)$.

$$F = x\overline{z} + \overline{w}\overline{z}$$

		CD			
		00	01	11	10
AB	00	1	0	0	X
	01	1	0	0	1
	11	1	0	0	1
	10	0	0	0	X

9. BCD, $F(w, x, y, z) = \sum m(4, 5, 6, 7, 8, 9)$, $d = (10, \dots, 15)$.

$$F = w + x$$

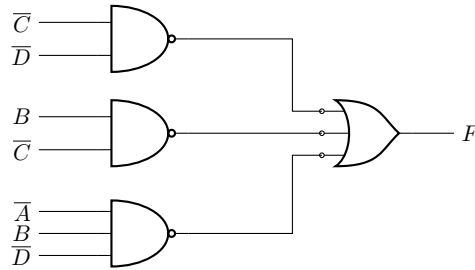
	CD			
	00	01	11	10
AB	00	0	0	0
	01	1	1	1
	11	X	X	X
	10	1	1	X

10. $F(A, B, C, D) = \sum m(0, 4, 5, 6, 8, 12, 13)$.

$$F = \overline{C}\overline{D} + B\overline{C} + \overline{A}B\overline{D}$$

		CD			
		00	01	11	10
AB	00	1	0	0	0
	01	1	1	0	1
	11	1	1	0	0
	10	1	0	0	0

NAND form (two faces of NAND: every AND $\rightarrow$ AND-INVERT, every OR $\rightarrow$ INVERT-OR):

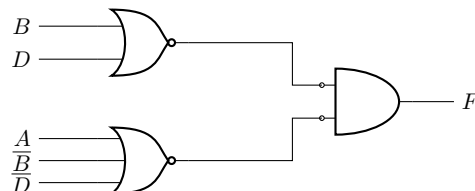


11. $F(A, B, C, D) = \prod M(0, 2, 5, 7, 8, 10)$. Grouping the 0's: $\overline{F} = \overline{B}\overline{D} + \overline{A}BD$, so

$$F = (B + D)(A + \overline{B} + \overline{D}).$$

		CD			
		00	01	11	10
AB	00	0	1	1	0
	01	1	0	0	1
	11	1	1	1	1
	10	0	1	1	0

NOR form (two faces of NOR: every OR $\rightarrow$ OR-INVERT, every AND $\rightarrow$ INVERT-AND):



12. (a)

$$\begin{aligned}
 x \oplus x \oplus y \oplus 0 &= (x \oplus x) \oplus y \oplus 0 && [\text{associativity}] \\
 &= 0 \oplus y \oplus 0 && [x \oplus x = 0] \\
 &= y && [a \oplus 0 = a]
 \end{aligned}$$

$$\boxed{y}$$

(b) Prove $\overline{x \oplus y} = x \oplus \bar{y}$.

$$\begin{aligned}
 \overline{x \oplus y} &= \overline{x\bar{y} + \bar{x}y} && [\text{definition of XOR}] \\
 &= \overline{(x\bar{y}) \cdot (\bar{x}y)} && [\text{DeMorgan}] \\
 &= (\bar{x} + y)(x + \bar{y}) && [\text{DeMorgan}] \\
 &= \bar{x}x + \bar{x}\bar{y} + xy + y\bar{y} && [\text{distribution}] \\
 &= \bar{x}\bar{y} + xy && [\bar{x}x = 0, y\bar{y} = 0] \\
 x \oplus \bar{y} &= x(\bar{y}) + \bar{x}y && [\text{definition of XOR}] \\
 &= xy + \bar{x}\bar{y} && [(\bar{y}) = y]
 \end{aligned}$$

Both reduce to $xy + \bar{x}\bar{y}$, so $\overline{x \oplus y} = x \oplus \bar{y}$. ■

(c) Prove $x \oplus (x \oplus y) = y$.

$$\begin{aligned}
 x \oplus (x \oplus y) &= (x \oplus x) \oplus y && [\text{associativity}] \\
 &= 0 \oplus y && [x \oplus x = 0] \\
 &= y && [0 \oplus a = a] \quad \blacksquare
 \end{aligned}$$

(d) $A \oplus B \oplus C = \sum m(1, 2, 4, 7)$.

Reason: a chain of XORs equals 1 exactly when an odd number of inputs are 1. The minterms with an odd count of 1's are 001, 010, 100, 111, i.e. 1, 2, 4, 7.

(e) $Equal = (A_4 \odot B_4)(A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1)(A_0 \odot B_0)$.

For $A = 11010$, $B = 11000$: every position matches except bit 1 ($A_1 = 1$, $B_1 = 0$), so $(A_1 \odot B_1) = 0$ and $Equal = 0$ — the numbers are **not equal**.

(f) Message $vwxyz = 10110$ has three 1's (odd). Odd parity requires an odd total, which already holds, so $P = 0$:

$$P = v \oplus w \oplus x \oplus y \oplus z = 0, \quad vwxyzP = \boxed{101100}.$$